Non-standard enumerable models of true arithmetic

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-model-theory-basics/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs: compactness, Downward LS, thm:isom (term-value part (a) only - that part is carried in full in the source; the formula-satisfaction part (b) of thm:isom is left as an exercise by the author and is NOT load-bearing here, only term values are used).

Editorial note. The label thm:non-std names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Compactness theorem. Same unit, compactness for first-order logic (standing result of the basics chapter). Finite satisfiability of N together with {num(n) < c} yields a model.
- Downward Loewenheim-Skolem. Same unit, Downward LS theorem (standing result). Trims the compactness model to an enumerable one.
- Term values under isomorphism. Same unit, \ollabel{thm:isom} proof part (a), term-value claim (source lines 580-613). Non-standardness contradiction: an isomorphism would carry the numeral n to the n-fold successor iterate of zero-interpretation, forcing c^M into the standard chain.

Source reference. Original theorem thm-isom: sources/olp-model-theory-basics/pinned-original-source.tex, source lines 575–575. Prerequisite pointer invoked at proof line(s) 702; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. *There are non-standard enumerable models of true arithmetic.*

Proof. Expand $\mathcal{L}L_N$ by introducing a new constant c, and consider the theory

$$\mathrm{Th}N \cup \{\, \overline{n} < c \mid n \in \mathbb{N} \,\}.$$

The theory is finitely satisfiable, so by compactness it has a model $\mathfrak{M}$, which can be taken to be enumerable by the Downward Löwenheim–Skolem theorem. Where $|\mathfrak{A}|M$ is the domain of $\mathfrak{M}$, let $\mathfrak{M}$ interpret the non-logical constants of $\mathcal{L}L$ as $\mathbf{z} = s0M \in |\mathfrak{A}|M$, $\prec = s{<}M \subseteq M^2$, $* = s\prime M \colon |\mathfrak{A}|M \to |\mathfrak{A}|M$, and $\oplus = s{+}M, \otimes = s{\times}M \colon |\mathfrak{A}|M^2 \to |\mathfrak{A}|M$. For each $x \in |\mathfrak{A}|M$, we write x^* for the element of $|\mathfrak{A}|M$ obtained from x by application of $*$.

Now, if h were an isomorphism of $\mathfrak{N}$ and $\mathfrak{M}$, there would be $n \in \mathbb{N}$ such that $h(n) = scM$. So let s be any assignment in $\mathfrak{N}$ such that $s(x) = n$. Then $\mathfrak{N}, s \vDash \overline{n} = x$; by the proof of thm-isom, also $\mathfrak{M}, h \circ s \vDash \overline{n} = x$, so that $scM = \mathbf{z}^{*\cdots *}$ (with $*$ iterated n times). But this is impossible since by assumption $\mathfrak{M} \vDash \overline{n} < c$ and $\prec$ is irreflexive. So $\mathfrak{M}$ is non-standard. □